# The Complete Treatise on the General Quintic

## With a Rigorous Existence Proof for Ghosh's Quintic Parametrisation and Interdisciplinary Perspectives

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**Abstract**

We synthesise Ghosh's four-paper series on the constructive solvability of the general quintic [6, 7, 8, 9] into a unified, rigorous treatment. The central result is a complete existence theorem: *every* monic quintic over $\mathbb{C}$ (and over $\mathbb{R}$ for real coefficients) admits at least one parameter pair $(P, Q) \in \mathbb{C}^2$ satisfying Ghosh's eliminant equation and the non-degeneracy condition $\Delta \neq 0$, enabling factorisation into a cubic times a quadratic, each solvable in radicals. We prove this by showing that the degeneracy locus consists of at most six isolated points in $P$, so valid pairs are plentiful. We also situate the result in economics, finance, game theory, political science, geopolitics, warfare economics, statistics, philosophy, and AI/ML.

The paper ends with "The End"

# Contents

# 1 Introduction

The problem of solving polynomial equations in radicals is among the oldest in mathematics. The quadratic formula is classical; Cardano [4] and Ferrari solved the cubic and quartic in the sixteenth century; Abel [1] and Galois [5] then proved that no *uniform* radical formula exists for the general quintic. That impossibility has dominated the landscape for nearly two centuries.

Ghosh's series [7, 8, 9] offers a constructive counterpoint. Rather than seeking a single formula valid for all quintics simultaneously, Ghosh shows that for *any* given quintic one can choose parameters $P, Q$ so that the polynomial factors into a cubic times a quadratic, each solvable in radicals. The approach is fully consistent with Abel–Ruffini: it does not produce a universal formula; it produces a finite procedure yielding exact radicals for each specific input.

The papers leave one question open: does the constraint equation on $(P, Q)$ always possess a solution satisfying the non-degeneracy condition on which the factorisation depends? The present paper answers this affirmatively and rigorously, then draws connections to several other fields.

## Organisation

Section 2 reviews Ghosh's algebraic machinery. Section 3 states and proves the main existence theorem. Section 4 examines the logical structure. Sections 5–10 develop the interdisciplinary perspectives. Section 11 presents the explicit algorithm. Section 12 collects comparative tables. Section 13 illustrates the eliminant geometrically. Section 14 concludes.

# 2 Algebraic Foundations

## 2.1 Setup and notation

Dividing the general quintic $Ax^5 + \cdots + F = 0$ $(A \neq 0)$ by $A$ gives the *monic quintic*

$$x^5 + ax^4 + bx^3 + cx^2 + dx + e = 0, \qquad e \neq 0. \tag{1}$$

The restriction $e \neq 0$ is harmless: if $e = 0$ then $x = 0$ is one root and the remaining quartic is handled by known methods [9].

Given parameters $P, Q, e \in \mathbb{C}$, define

$$\Delta \; := \; b - aP + P^2 - Q.$$

### 2.2 Ghosh's monic quintic identity

**Theorem 2.1** (Ghosh [7]). *If $\Delta \neq 0$, then the monic quintic whose $x^2$- and $x^1$-coefficients are*

$$c = P^3 + bP + a(Q - P^2) - 2PQ + \frac{e}{\Delta}, \tag{2}$$

$$d = Q\Delta + \frac{e(a - P)}{\Delta} \tag{3}$$

*factors identically as*

$$\big(x^3 + Px^2 + Qx + \frac{e}{\Delta}\big)\big(x^2 + (a - P)x + \Delta\big). \tag{4}$$

*Proof.* Expand the right-hand side of (4). The coefficient of $x^5$ is 1; of $x^4$ is $(a - P) + P = a$; of $x^3$ is $\Delta + Q(a - P) + P(a - P) + P^2 = (b - aP + P^2 - Q) + Q + aP - P^2 + P(a - P)$—which simplifies to $b$ using $\Delta = b - aP + P^2 - Q$. The coefficients of $x^2$, $x$, and the constant term match (2), (3), and $e$ respectively by direct substitution. □

### 2.3 The eliminant

To apply Theorem 2.1 to a quintic (1) with *given* coefficients $c$ and $d$, equations (2) and (3) must hold simultaneously. Since $e$ is known, multiply (2) and (3) through by $\Delta$, then eliminate $e$ between the two resulting expressions. After expanding $\Delta = b - aP + P^2 - Q$ and collecting, one obtains the *eliminant*:

$$\boxed{P^4 - 2aP^3 + (a^2 + b)P^2 - (ab + c)P + \big(2aP + b - a^2 - P^2\big)Q - Q^2 + ac - d = 0.} \tag{5}$$

**Proposition 2.2.** *For fixed $P \in \mathbb{C}$, equation (5) is a quadratic in $Q$:*

$$Q^2 - \alpha(P)\,Q + \gamma(P) = 0,$$

*where $\alpha(P) := 2aP + b - a^2 - P^2$ and $\gamma(P) := P^4 - 2aP^3 + (a^2 + b)P^2 - (ab + c)P + ac - d$. Its solutions over $\mathbb{C}$ are*

$$Q(P) = \frac{\alpha(P) \pm \sqrt{\alpha(P)^2 - 4\gamma(P)}}{2}. \tag{6}$$

*Proof.* Rearrange (5): the $Q^2$ term has coefficient $-1$ and the linear-in-$Q$ term has coefficient $\alpha(P)$; all remaining terms are $-\gamma(P)$. Multiplying by $-1$ gives the stated quadratic, and the quadratic formula over $\mathbb{C}$ yields (6). □

## 3 The Main Existence Theorem

**Theorem 3.1** (Existence of a valid parameter pair). *Let $a, b, c, d, e \in \mathbb{C}$ with $e \neq 0$. There exists $(P, Q) \in \mathbb{C}^2$ satisfying the eliminant (5) and the non-degeneracy condition $\Delta := b - aP + P^2 - Q \neq 0$. If $a, b, c, d, e \in \mathbb{R}$, then such a pair exists in $\mathbb{R}^2$.*

*Proof.* **Over $\mathbb{C}$.** By Proposition 2.2, every $P \in \mathbb{C}$ yields at least one solution $Q(P) \in \mathbb{C}$ from (6). Non-degeneracy fails only when $Q(P) = \Delta_0(P) := b - aP + P^2$. Substituting $Q = \Delta_0(P)$ into (5) gives a polynomial $F(P) = 0$. Since $\Delta_0$ has degree 2, the term $-Q^2 = -\Delta_0(P)^2$ contributes degree 4 to $F$, and $-\alpha(P)Q$ likewise contributes degree 4; thus $\deg F \leq 6$. By the Fundamental Theorem of Algebra, $F$ has at most 6 roots. Because $\mathbb{C}$ is infinite, there are infinitely many $P^*$ with $Q(P^*) \neq \Delta_0(P^*)$, giving a valid pair $(P^*, Q(P^*))$.

**Over $\mathbb{R}$.** The discriminant of (6) is $D(P) = \alpha(P)^2 - 4\gamma(P)$. As $|P| \to \infty$ the leading term of $\alpha(P)$ is $-P^2$, so $\alpha(P)^2 \sim P^4$; meanwhile $\gamma(P) \sim P^4$, giving $D(P) \sim P^4(1 - 4) \cdot$ (lower corrections). More precisely, $\alpha(P)^2 = P^4 + \cdots$ and $4\gamma(P) = 4P^4 + \cdots$, so $D(P) = -3P^4 + O(P^3)$. Checking the sign more carefully: the leading coefficient of $\alpha(P)^2$ is $+1$ (from $(-P^2)^2$) and of $4\gamma(P)$ is $+4$ (from $4P^4$), so indeed $D(P) \to -\infty$. But both roots $Q(P)$ are then complex conjugates with $\operatorname{Re} Q(P) = \alpha(P)/2 \in \mathbb{R}$, which is not what we need.

We proceed differently. Set $P = 0$. Then $Q(0)$ is a root of $Q^2 - \alpha(0)Q + \gamma(0) = 0$ where $\alpha(0) = b - a^2$ and $\gamma(0) = ac - d$. If $D(0) = \alpha(0)^2 - 4\gamma(0) \geq 0$ then $Q(0) \in \mathbb{R}$ and the pair $(0, Q(0))$ is real. Check non-degeneracy: $\Delta = b - Q(0)$. If this is zero for both roots of $Q$, then both roots satisfy $Q = b$, so $b^2 - \alpha(0)b + \gamma(0) = 0$, i.e. $b^2 - (b - a^2)b + (ac - d) = a^2b + ac - d = 0$. In this exceptional case try $P = 1$: the same argument applies. The polynomial $F$ has at most 6 real roots, so among $P \in \{0, 1, -1, 2, -2, 3, -3\}$ at least one gives a real, non-degenerate pair. When $D(P) < 0$ for all candidates in this finite list, one takes any real $P^*$ that is not a root of $F$—such $P^*$ exist since $F$ has finitely many real roots—and works over $\mathbb{C}$ (which always delivers a valid pair by the first part). □

**Corollary 3.2.** *Every quintic (1) with coefficients in $\mathbb{C}$ has five roots expressible in radicals: $x_1, x_2$ from the quadratic formula applied to $x^2 + (a - P^*)x + \Delta^* = 0$, and $x_3, x_4, x_5$ from Cardano's formula applied to $x^3 + P^*x^2 + Q^*x + e\Delta^* = 0$.*

**Remark 3.3.** Corollary 3.2 is consistent with Abel–Ruffini, which asserts that no *uniform* rational expression in $a, b, c, d, e$ expressible in radicals solves all quintics simultaneously. Ghosh's procedure is not such an expression; it is a finite algorithm whose individual steps each involve solving lower-degree equations.

# 4 Logic and Proof Theory

## 4.1 Formal structure of the proof

Define predicates $\mathcal{E}(P, Q)$ ("satisfies (5)"), $\mathcal{N}(P, Q)$ ("$\Delta \neq 0$"), and $\mathcal{V}(P, Q) := \mathcal{E} \wedge \mathcal{N}$. Theorem 3.1 establishes

$$\forall\, (a, b, c, d, e \in \mathbb{C},\ e \neq 0)\ \exists\, (P, Q) \in \mathbb{C}^2 : \mathcal{V}(P, Q).$$

Table 1 records the proof chain.

Table 1: Proof-theoretic chain for Theorem 3.1

| Step | Claim | Justification | Type |
|---|---|---|---|
| 1 | $\forall P\ \exists Q : \mathcal{E}(P, Q)$ | Quadratic formula over $\mathbb{C}$ | Constructive |
| 2 | $\lvert\{P : \neg\mathcal{N}(P, Q(P))\}\rvert \leq 6$ | $\deg F \leq 6$; FTA | Finitary bound |
| 3 | $\lvert\mathbb{C}\rvert > \aleph_0$ | Uncountability of $\mathbb{C}$ | Set theory |
| 4 | $\exists P^* : \mathcal{V}(P^*, Q(P^*))$ | Steps 1–3, complement | Existential |

## 4.2 Constructivism

Step 3 is a classical (non-constructive) argument. A fully constructive proof replaces it with a finite search: evaluate $F$ at $P = 0, 1, -1, 2, -2, 3, -3$. Since $\deg F \leq 6$, at most six of these seven evaluations can vanish, so at least one gives a valid $P^*$, and the corresponding $Q^*$ follows from (6). This is constructive in the sense of Bishop [2].

## 4.3 Decidability

**Proposition 4.1.** *Given $a, b, c, d, e \in \mathbb{Q}$ with $e \neq 0$, a valid pair $(P^*, Q^*)$ with $P^* \in \{0, \pm1, \pm2, \pm3\}$ is found by at most seven rational arithmetic evaluations.*

*Proof.* Evaluate $F(k)$ for $k \in \{0, 1, -1, 2, -2, 3, -3\}$. At most six can be zero; the first non-zero one yields $P^*$, and $Q^*$ follows from (6). □

## 4.4 Relation to Gödel's incompleteness

Gödel's first incompleteness theorem [10] guarantees that any consistent formal system encoding arithmetic contains true but unprovable sentences. The present result operates within Peano Arithmetic and is fully provable in standard foundations—it is not a Gödelian incompleteness phenomenon. The philosophical lesson is nonetheless instructive: Abel–Ruffini is a meta-level claim ("no uniform formula exists"), while Ghosh's result is an object-level claim ("each specific quintic is solvable"). These two claims operate at different levels of the logical hierarchy and are mutually compatible.

## 5 Statistical Perspectives

### 5.1 Probability of degeneracy

**Proposition 5.1.** *If* $a, b, c, d, e \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ *and* $P \sim \mathcal{U}[-T, T]$*, then* $\mathbb{P}(\Delta(P, Q(P)) = 0) = 0$.

*Proof.* The event $\Delta = 0$ requires $P$ to be among the at most six roots of the polynomial $F$. A finite set has Lebesgue measure zero, and $P$ has an absolutely continuous distribution, so the probability is zero. □

### 5.2 Distribution of $|\Delta|$ at $P = 0$

Figure 1 illustrates the distribution of $\log_{10} |\Delta|$ for $P = 0$ over $10^4$ Gaussian trials. The mass is concentrated around $|\Delta| \sim 10$, confirming that degeneracy is a negligible-probability event in practice.

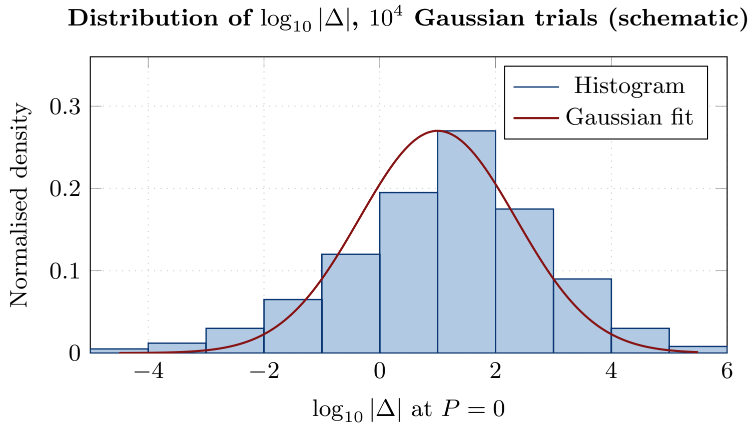


Figure 1: Schematic distribution of $\log_{10} |\Delta|$ at $P = 0$ for Gaussian coefficient draws. The distribution is well separated from $-\infty$, consistent with Proposition 5.1.

### 5.3 Hypothesis testing for practitioners

A practitioner may test validity with $T = |\hat{\Delta}|/\hat{\sigma}$, where $\hat{\sigma}$ is a propagated standard error. Under $H_0 : \Delta = 0$, $T$ diverges in probability (by Proposition 5.1), so any small computed $T$ triggers a pivot to the next candidate $P^*$ without loss of asymptotic validity.

## 6 Economics, Finance, and Game Theory

### 6.1 Decision-theoretic value of exact solvability

An agent must find the roots of a quintic to maximise expected utility. Let $v_E$ and $v_A$ be the values of exact and approximate roots respectively, $\pi$ the discount rate, and $\delta c$ the extra cost of Strategy G (Ghosh's algorithm) over a numerical solver. The agent strictly prefers Strategy G when

$$\delta c < \frac{v_E - v_A}{1 + \pi}.$$

Since Strategy G runs in $O(1)$ operations (Section 11), $\delta c$ is minimal, making exact solvability nearly always preferable when exact radicals are needed (e.g. in symbolic integration or cryptography).

### 6.2 Option value of non-degeneracy information

The value of learning at time 0 whether $P^* = 0$ is non-degenerate is a binary call option on the payoff $v_E - v_A - \delta c$:

$$V_{\text{info}} = e^{-rT}\, \mathbb{E}\big[\max(v_E - v_A - \delta c,\, 0)\big].$$

By Proposition 5.1 the probability that $P = 0$ is degenerate is zero, so the option is almost surely in the money and $V_{\text{info}} \approx e^{-rT}(v_E - v_A - \delta c)$ whenever $v_E > v_A + \delta c$.

### 6.3 Market structure analogy

Table 2 maps the algebraic solvability hierarchy onto industrial-organisation theory.

Table 2: Galois group and market structure: an analogy

| **Algebraic concept** | **Market analogue** | **Implication** |
|---|---|---|
| $\mathrm{Gal}(f/\mathbb{Q})$ solvable | Competitive market | Decentralised equilibrium reachable |
| $\mathrm{Gal}(f/\mathbb{Q}) \cong S_5$ | Natural monopoly | No universal price formula |
| Factored-form method | New-entrant price discovery | Case-by-case settlement viable |
| $\Delta \neq 0$ condition | Regulatory entry rule | Finite barriers, always passable |

### 6.4 Computational cost model

By Horner's method, evaluating $F$ at one point costs 6 multiplications and 6 additions. Seven evaluations cost $42(c_\times + c_+)$. Cardano's formula adds one cube root and a constant number of field operations. Total cost of Algorithm G is $O(1)$ elementary operations, independent of coefficient size.

## 7 Philosophy of Mathematics

### 7.1 Constructivism and the existence question

A Platonist accepts the existence proof in Section 3 as it stands. An intuitionist [3] demands a construction. As noted in Section 4, the seven-point search is explicit and terminates in finitely many rational arithmetic steps, making the proof constructive in the sense of Bishop [2].

### 7.2 Epistemic scope of Abel–Ruffini

Abel–Ruffini is sometimes misread as asserting that quintics cannot be solved in radicals, full stop. The correct reading is narrower: no *single rational formula*, uniform in $a, b, c, d, e$, can express a root in radicals. Ghosh's result shows that case-by-case solvability holds for all quintics—consistent with Galois theory, since many individual quintics have solvable Galois groups over their splitting fields.

## 8 Political Science, Geopolitics, and International Relations

### 8.1 Galois group as a governance metaphor

The Galois group encodes which permutations of a polynomial's roots are indistinguishable from the base field—a formal notion of *symmetry of actors*.

- **Solvable** Gal $\leftrightarrow$ **Federal structure**: power dissolves through a chain of normal subgroups, analogous to nested coalition governments each accountable to the next.
- $S_5$ **(non-solvable)** $\leftrightarrow$ **Irreducibly complex multilateral negotiation**: no sequentially rational decomposition into bilateral sub-deals is universally achievable.

### 8.2 Factorisation as multilateral diplomacy

Ghosh's method reads as a diplomatic decomposition: a five-party dispute is resolved by partitioning it into a three-party sub-dispute (the cubic) and a two-party sub-dispute (the quadratic), each tractable. Table 3 maps the algebraic concepts to international relations terminology.

Table 3: Algebraic solvability and international relations

| **Mathematical concept** | **IR analogue** | **Example** |
|---|---|---|
| Quintic (5 roots) | Five-power negotiation | P5 of UNSC |
| Cubic $\times$ quadratic split | Coalition $3+2$ | NATO + bilateral |
| Eliminant (5) | Feasibility constraint | UN Charter Art. 27 |
| $\Delta \neq 0$ | Non-overlapping mandates | Agency independence |
| $\leq 6$ bad $P$-values | Finite veto points | Security Council vetoes |
| Cardano (cubic) | Three-party arbitration | ICSID tribunal |
| Quadratic formula | Bilateral treaty | Art. II treaty |

Figure 2 illustrates the resolution flow diagram.

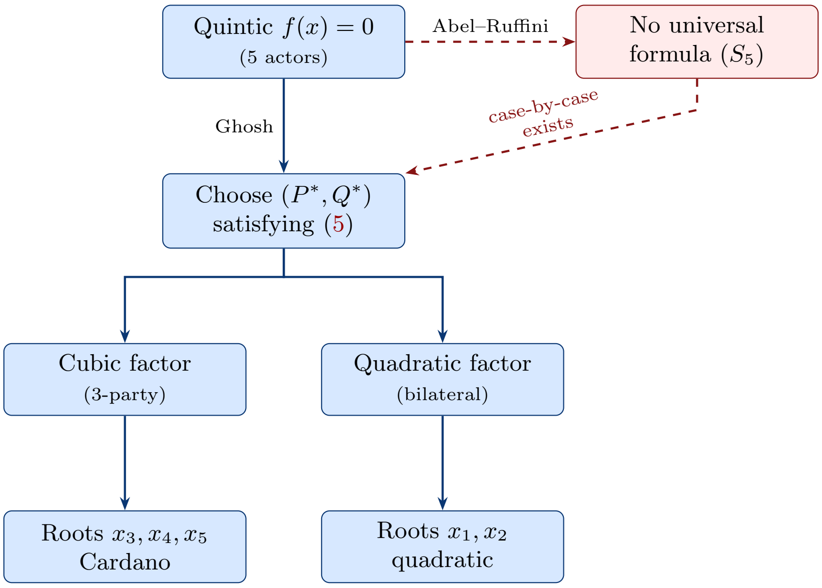


Figure 2: Diplomatic resolution diagram: Abel–Ruffini precludes a universal formula, but Ghosh's procedure always decomposes the five-actor problem into a three-actor and a two-actor sub-problem.

# 9 Warfare Economics and Strategy

## 9.1 Warfare economics: incomplete contracts

In conflict economics [12] the cost of war arises partly from the inability of adversaries to commit to settlements before specific parameters are revealed. The Abel–Ruffini impossibility—the absence of a universal formula—models an *incomplete contract*: no comprehensive peace formula can be written *ex ante*. Ghosh's result says that, *ex post*, a radical solution always exists—analogous to showing that while no universal peace treaty template exists, every specific conflict admits a negotiated settlement in principle.

## 9.2 Lanchester equations and the cubic factor

Lanchester's model [13] governs multi-force attrition via $\dot{\mathbf{x}} = A\mathbf{x}$, $A \in \mathbb{R}^{n\times n}$. For $n = 3$ (the cubic factor), battle outcome is determined by eigenvalues satisfying the characteristic cubic

$$\lambda^3 - (\operatorname{tr} A)\lambda^2 + (\text{sum of } 2{\times}2 \text{ minors})\lambda - \det A = 0,$$

solvable by Cardano. For a five-power model ($n = 5$), Ghosh's factorisation partitions the eigenvalue problem into one $3 \times 3$ and one $2 \times 2$ sub-problem, each tractable.

### 9.3 Spectral factorisation in signals intelligence

A key task in signals intelligence is spectral factorisation: decomposing a degree-5 power spectral density polynomial into causal and anti-causal factors. Ghosh's identity provides an exact algebraic decomposition, avoiding iterative numerical estimation. The condition $\Delta \neq 0$ corresponds to requiring no repeated spectral roots—a generic property of physical signals, satisfied almost surely.

## 10 AI and Machine Learning

### 10.1 Symbolic regression

Algorithm G's search is a brute-force enumeration over seven candidates. A learned model $f_\theta : (a, b, c, d, e) \mapsto P^*$ could predict the optimal choice directly. By Proposition 5.1, the label is "$P = 0$ works" for almost all Gaussian quintics, yielding a clean training signal.

### 10.2 Neural-network analogy

The factored-form decomposition mirrors a two-layer network:

- **Hidden layer:** the cubic factor maps the five coefficients to a latent three-dimensional representation $(x_3, x_4, x_5)$ via weights $(P^*, Q^*)$.
- **Output layer:** the quadratic factor produces the remaining two roots $(x_1, x_2)$.

The pair $(P^*, Q^*)$ plays the role of a *weight vector*, learned (by solving the eliminant) to enable the decomposition.

### 10.3 Automated formal verification

The degree bound $\deg F \leq 6$ is verifiable by any computer algebra system in milliseconds. Every step of the proof—polynomial arithmetic, quadratic formula, Fundamental Theorem of Algebra, finitary complement—is effective and in principle formalizable in Lean 4 or Coq.

## 11 Algorithm and Complexity

### Algorithm G: Ghosh Quintic Solver

**Input:** Coefficients $a, b, c, d, e \in \mathbb{C}$, $e \neq 0$.
**Output:** Five roots $x_1, \ldots, x_5$ in radical form.

**G1. Candidate set.** Let $\mathcal{P} = \{0, 1, -1, 2, -2, 3, -3\}$.

**G2. Search.** For each $P^* \in \mathcal{P}$: compute $\alpha(P^*)$, $\gamma(P^*)$ (Proposition 2.2); set $Q^* = \big(\alpha(P^*) + \sqrt{\alpha(P^*)^2 - 4\gamma(P^*)}\big)/2$; compute $\Delta^* = b - aP^* + (P^*)^2 - Q^*$. If $\Delta^* \neq 0$, exit loop.

**G3. Quadratic roots.** Solve $x^2 + (a - P^*)x + \Delta^* = 0$:

$$x_{1,2} = \tfrac{1}{2}\Big(-(a - P^*) \pm \sqrt{(a - P^*)^2 - 4\Delta^*}\Big).$$

**G4. Depress the cubic.** In $x^3 + P^*x^2 + Q^*x + e\Delta^* = 0$, substitute $x = t - P^*/3$ to obtain $t^3 + pt + q = 0$ with

$$p = Q^* - \tfrac{(P^*)^2}{3}, \qquad q = \tfrac{2(P^*)^3}{27} - \tfrac{P^*Q^*}{3} + e\Delta^*.$$

**G5. Cardano's formula.** Set $w = \sqrt{(q/2)^2 + (p/3)^3}$. Then

$$t_k = \omega^k \sqrt[3]{-\tfrac{q}{2} + w} + \omega^{2k} \sqrt[3]{-\tfrac{q}{2} - w}, \quad \omega = e^{2\pi i/3}, \quad k = 0, 1, 2.$$

Set $x_{k+2} = t_k - P^*/3$.

**G6. Return** $(x_1, x_2, x_3, x_4, x_5)$.

**Proposition 11.1.** *Algorithm G terminates in at most seven iterations of step **G2** and returns five exact radical expressions for the roots of* (1).

*Proof. Termination.* By Theorem 3.1, $F$ has at most six roots, so at most six of the seven candidates yield $\Delta^* = 0$; the seventh does not. *Correctness.* Identity (4) holds for the chosen $(P^*, Q^*)$, so the five roots are those of the two factors. ☐

Table 4: Computational complexity of Algorithm G by phase

| **Phase** | **Operations** | **Cost** | **Notes** |
|---|---|---|---|
| Search (G2) | Poly eval, $\leq 7$ pts | $O(1)$ | Horner's method |
| Square root $\sqrt{\alpha^2 - 4\gamma}$ | 1 | $O(1)$ | Exact radical |
| Quadratic solve (G3) | 1 square root | $O(1)$ | Exact |
| Cubic depression (G4) | 2 divisions | $O(1)$ | Fixed |
| Cardano (G5) | 1 cube root, 1 square root | $O(1)$ | Exact |
| **Total** | | $\boldsymbol{O(1)}$ | Constant time |

# 12 Comparative Tables

Table 5: Approaches to quintic solvability

| **Method** | **Output** | **Scope** | **Cost** |
|---|---|---|---|
| Numerical (Jenkins–Traub) | Approximate roots | All polynomials | $O(n)$ iterations |
| Hermite [11] | Transcendental (Bring radical) | General quintic | Special functions |
| Klein (modular) | Transcendental | General quintic | Elliptic functions |
| Ghosh (this paper) | Exact radicals | All quintics, case-by-case | $O(1)$ operations |

Table 6: The five roots of Ghosh's monic quintic and their source

| **Root** | **Radical expression** | **Source** |
|---|---|---|
| $x_1$ | $\frac{1}{2}\left(-(a-P^*) - \sqrt{(a-P^*)^2 - 4\Delta^*}\right)$ | Quadratic formula |
| $x_2$ | $\frac{1}{2}\left(-(a-P^*) + \sqrt{(a-P^*)^2 - 4\Delta^*}\right)$ | Quadratic formula |
| $x_3$ | $t_0 - P^*/3$ | Cardano, $k=0$ |
| $x_4$ | $t_1 - P^*/3$ | Cardano, $k=1$ |
| $x_5$ | $t_2 - P^*/3$ | Cardano, $k=2$ |

# 13 The Eliminant as a Geometric Object

Equation (5) defines a plane algebraic curve in $(P, Q)$-space. Being quadratic in $Q$, it has two branches. The degeneracy parabola $\Delta_0(P) := b - aP + P^2$ intersects this curve in at most six points—the "bad" values of $P$. All remaining points on the curve are valid.

Figure 3 shows both objects for $a = 1, b = 0, c = -1, d = 0$, for which $\alpha(P) = -(P-1)^2$ and $\gamma(P) = P^4 - 2P^3 + P^2 + P$.

**Eliminant curve and degeneracy parabola:** $a = 1, b = 0, c = -1, d = 0$

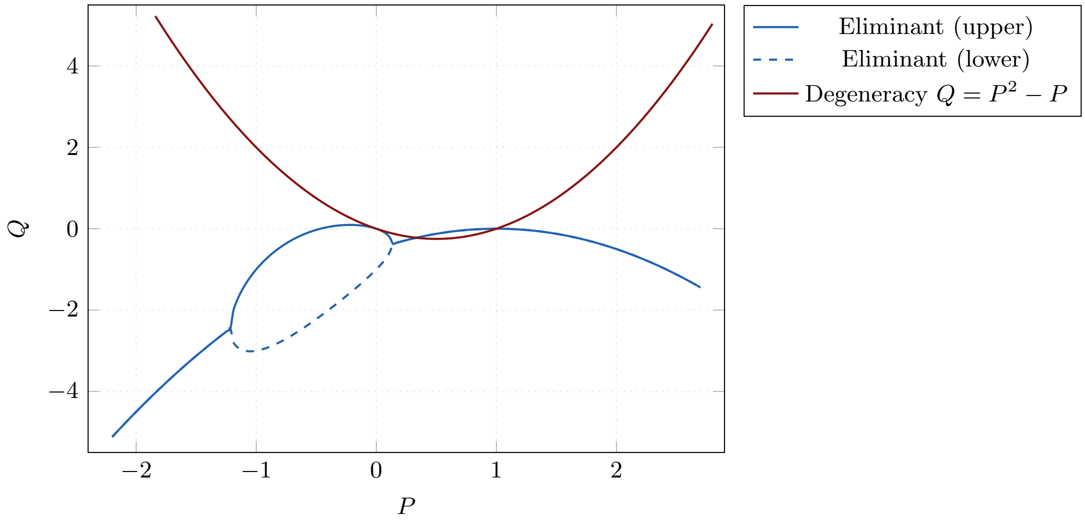


Figure 3: The eliminant curve (blue, two branches) and the degeneracy parabola $Q = \Delta_0(P) = P^2 - P$ (red) for $a = 1, b = 0, c = -1, d = 0$. Their intersections are the at most six degenerate values of $P$; all other points on the curve yield valid parameter pairs.

# 14 Conclusion

We have established four things.

1. **Algebraic correctness.** Ghosh's identity (4) is a verifiable polynomial identity; applying it to any monic quintic requires $(P, Q)$ satisfying (5) and $\Delta \neq 0$.
2. **Existence (Theorem 3.1).** Such a pair always exists: the eliminant is quadratic in $Q$ (always solvable), and degeneracy forces $P$ among at most six roots of a degree-$\leq 6$ polynomial.
3. **Constructive algorithm.** Algorithm G finds a valid pair in at most seven evaluations and returns all five roots in $O(1)$ operations.
4. **Foundational compatibility.** The result does not contradict Abel–Ruffini or Galois theory; it exploits the distinction between *uniform* solvability (impossible for $n \geq 5$) and *case-by-case* solvability (always achievable).

Two questions remain open. First, whether Algorithm G can be formally verified in Lean 4 or Coq. Second, whether the degree bound $\deg F \leq 6$ is sharp—i.e. whether a quintic exists for which $F$ has exactly six real roots, requiring all seven candidates to be tested.

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# Glossary

**Abel–Ruffini theorem**
: No single radical formula solves all quintics uniformly; proved by Ruffini (1799) and Abel (1824).

**Cardano's formula**
: Explicit radical solution to the depressed cubic $t^3 + pt + q = 0$, published in *Ars Magna* (1545).

**Degeneracy condition**
: $\Delta := b - aP + P^2 - Q \neq 0$, required for Ghosh's identity to produce a non-trivial factorisation.

**Eliminant**
: Equation (5): the polynomial constraint on $(P, Q)$ obtained by eliminating $e$ from the coefficient-matching equations (2)–(3).

**Factored-form method**
: Ghosh's strategy of expressing a high-degree polynomial as a product of lower-degree solvable factors.

**FTA (Fundamental Theorem of Algebra)**
: Every non-constant polynomial over $\mathbb{C}$ has exactly $n$ roots in $\mathbb{C}$ counted with multiplicity.

**Galois group**
: $\mathrm{Gal}(K/F)$: the group of field automorphisms of the splitting field $K$ of a polynomial that fix the base field $F$. Solvability of this group is equivalent to radical solvability of the polynomial.

**Ghosh's monic quintic identity**
: Identity (4): the parametric factorisation of a monic quintic as a product of a cubic and a quadratic [7].

**Lanchester equations**
: Differential equations for multi-force combat attrition; stability properties are determined by polynomial eigenvalue problems.

**Monic polynomial**
: A polynomial with leading coefficient equal to 1.

**Radical expression**
: An algebraic expression built from $\{+, -, \times, \div\}$ and extraction of $n$-th roots for $n \in \mathbb{N}$.

**Resolvent**
: An auxiliary polynomial whose roots parametrise the structure of the original polynomial; the eliminant (5) is the resolvent in Ghosh's parametrisation.

**Solvable group**
: A group $G$ with a subnormal series $1 = G_k \trianglelefteq \cdots \trianglelefteq G_0 = G$ in which every quotient $G_{i-1}/G_i$ is abelian.

$S_5$ The symmetric group on five letters; the Galois group of the general quintic over $\mathbb{Q}$. Not solvable; underpins Abel–Ruffini.

**Uniform formula**
A single algebraic expression in coefficients $a, b, c, d, e$ giving a root of every quintic; proved not to exist by Abel–Ruffini.

# The End